Earthquake Hazard and Exposure Analysis Using R and GIS

Project Overview

This project presents a simplified earthquake hazard and exposure analysis using R and GIS. The workflow covers data preparation, exploratory hazard analysis, spatial processing and identification of populated areas potentially exposed to recorded earthquake events. The project demonstrates elements of the hazard and exposure stages of catastrophe risk modelling, while vulnerability, financial loss and probabilistic modelling are outside the scope of the analysis.

Datasets

- Dataset 1- Disasters: contains a list of natural disasters categories by both hazard type and year (over a 40 year period between 1980 and 2020), offering a broad context of hazards.
- Dataset 2- Earthquakes: individual earthquake events over a brief time period between September to October 2022, includes information on location, magnitude and depth.
- Dataset 3- Countries: contains the geographic borders of countries which are used to aggregate and identify earthquakes by country.
- Dataset 4- Global Human Settlement: consists of the locations of cities or populated settlements.

Methodology

The datasets were imported into R and underwent cleaning and validation procedures. Analysis was undertaken to assess disaster frequency and temporal trends, followed by earthquake magnitude and depth. Subsequently earthquake coordinates were converted into spatial features and mapped alongside country boundaries. Finally a 50km buffer was applied to the epicentre of each earthquake event to identify populated settlements potentially exposed to the events.

Results

Besides extreme weather and flooding disasters, which have substantially the most events per annum, earthquake events are the next most prevalent disaster category (Figure 1). The overall picture shown by Figure 2 is that the number of disaster events is rising. Most evident is the sharp rise in the number of events during the latter 1990s. Then, since the year 2000 the number of disasters has remained high, although the overall trend appears to be a slight decline in disaster events. It is evident that there are far more extreme weather and flooding events than wildfires, droughts or extreme temperature events (Figure 3), as a result, Figures 3a and 3b plot these separately to more clearly display the trends over time. There is a substantial rise in the number of flooding events, especially during the mid 1990s and 2000s. There is a far more gradual and less pronounced increase in extreme weather events. Otherwise, most of the climatic disasters have remained relatively stable over the 40 year period, with the exception of a minor, gradual rise in extreme temperature events (Figures 3a&3b). There are no identifiable trends in the number of non-climatic hazard events (Figure 4).

Figure 1- Frequency of Disasters

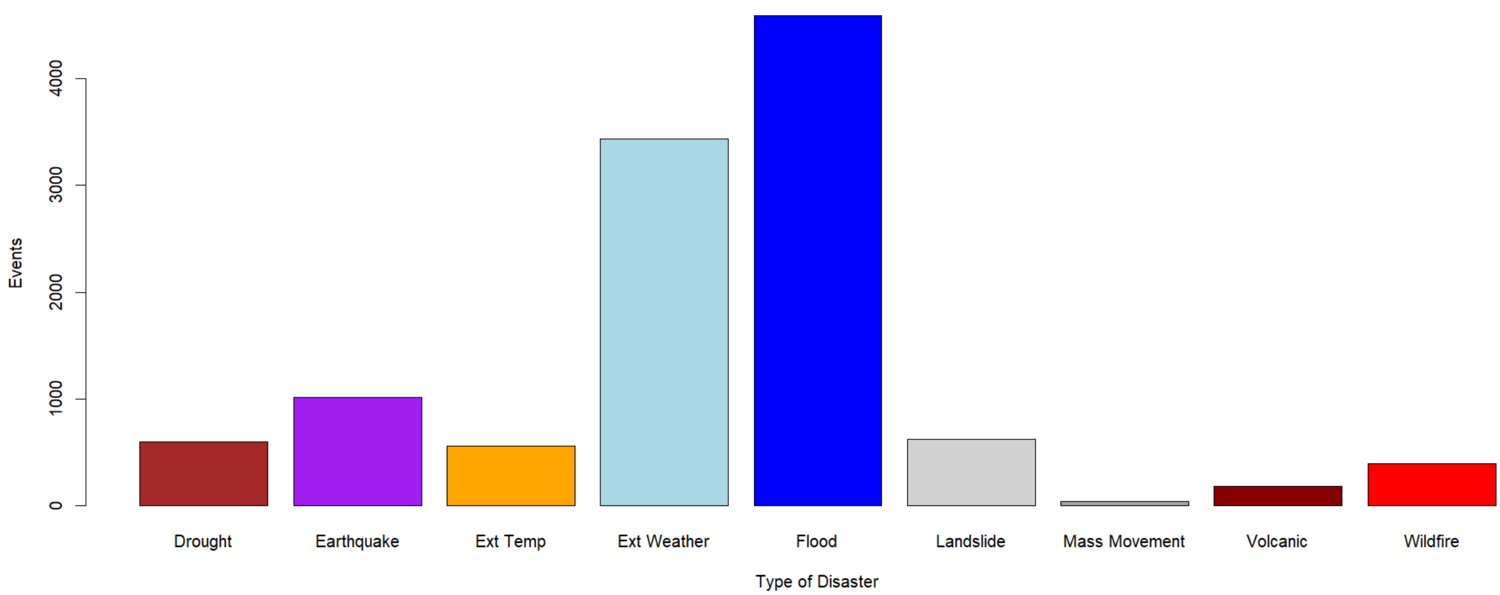


Figure 2- Global Disasters Timeline

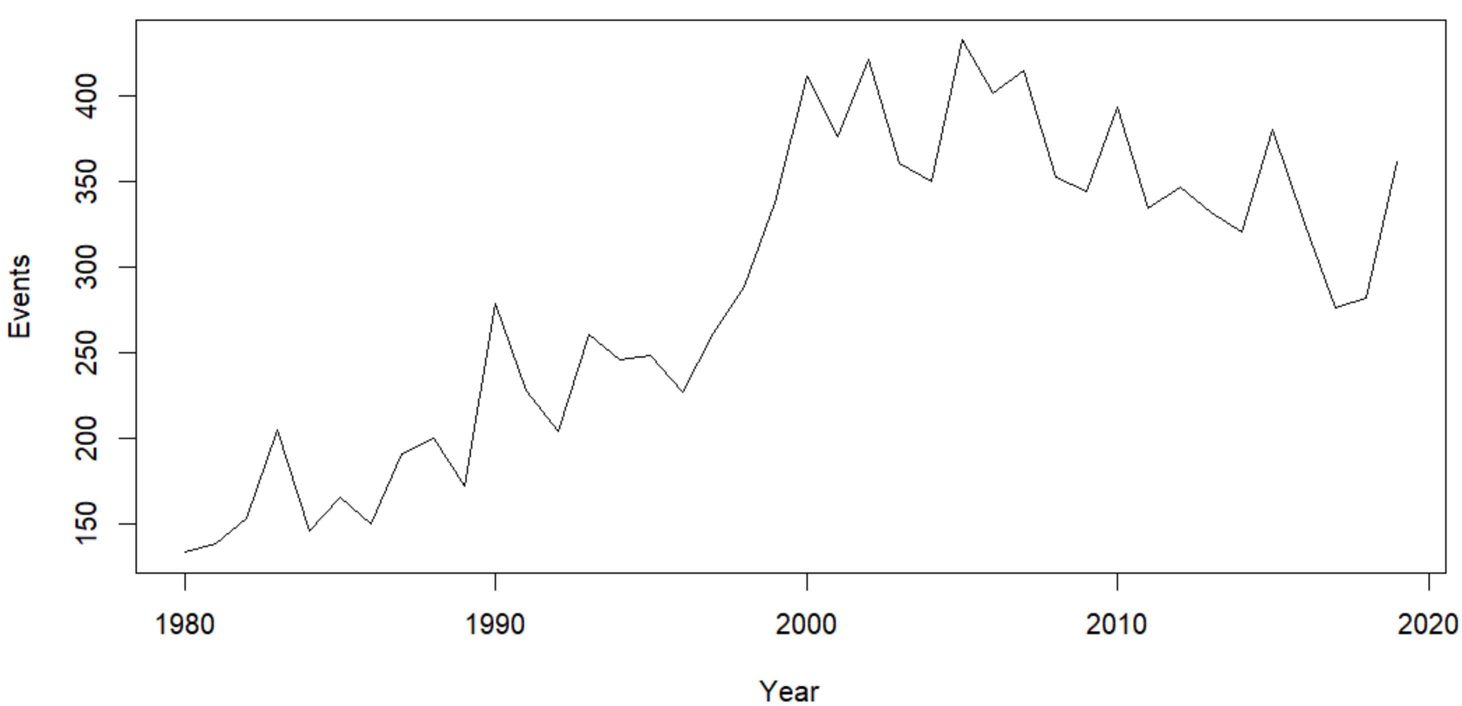


Figure 3- Global Disasters Timeline (Climatic)

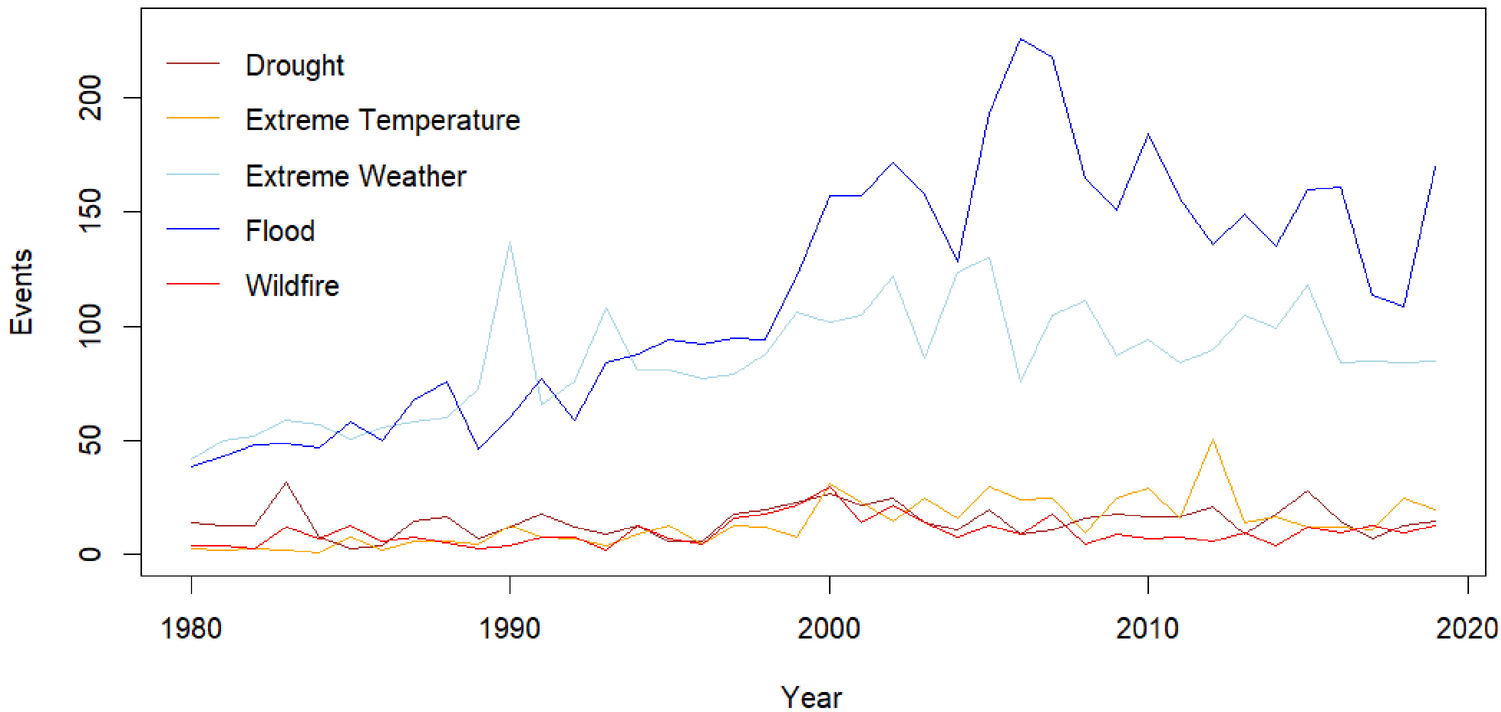
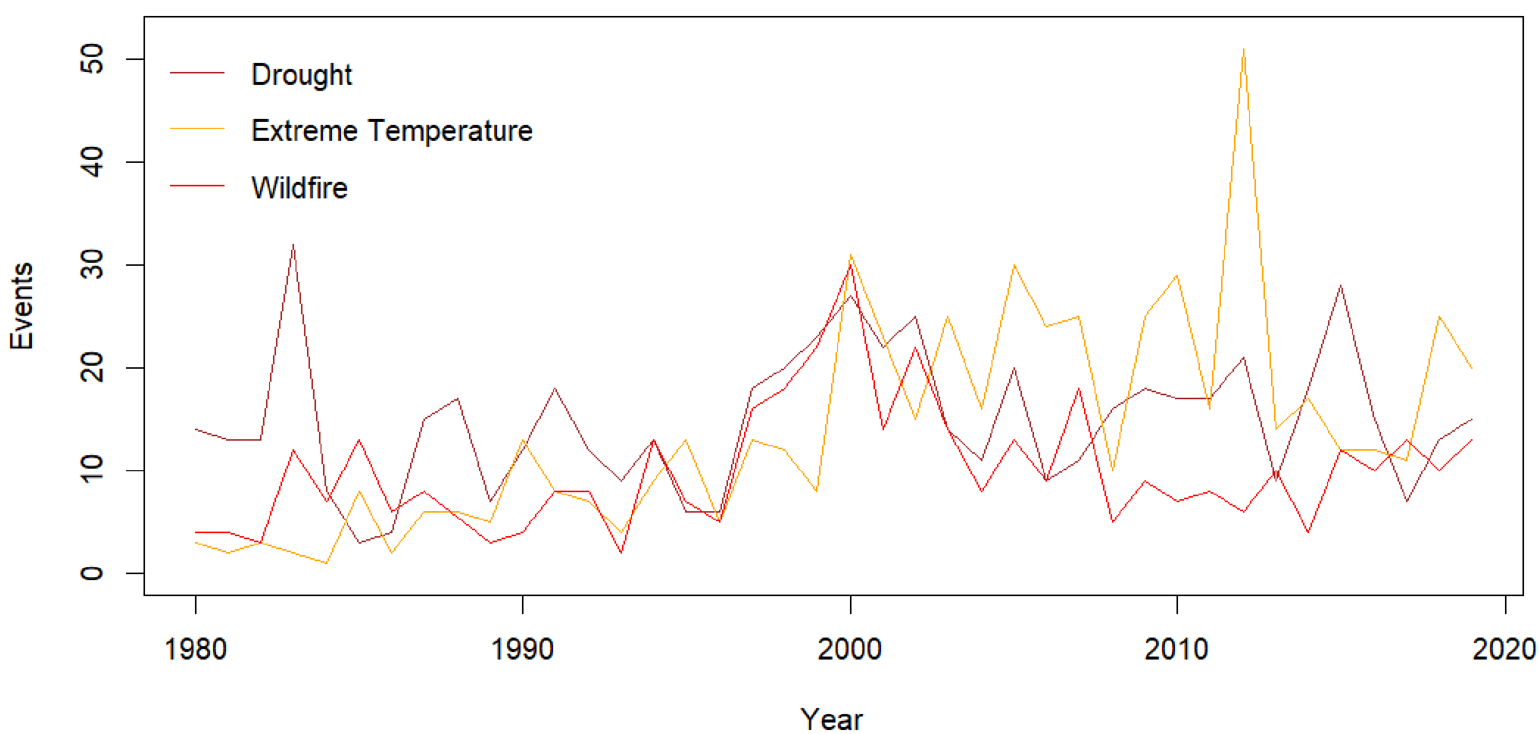


Figure 3a- Global Disasters Timeline (Climatic)



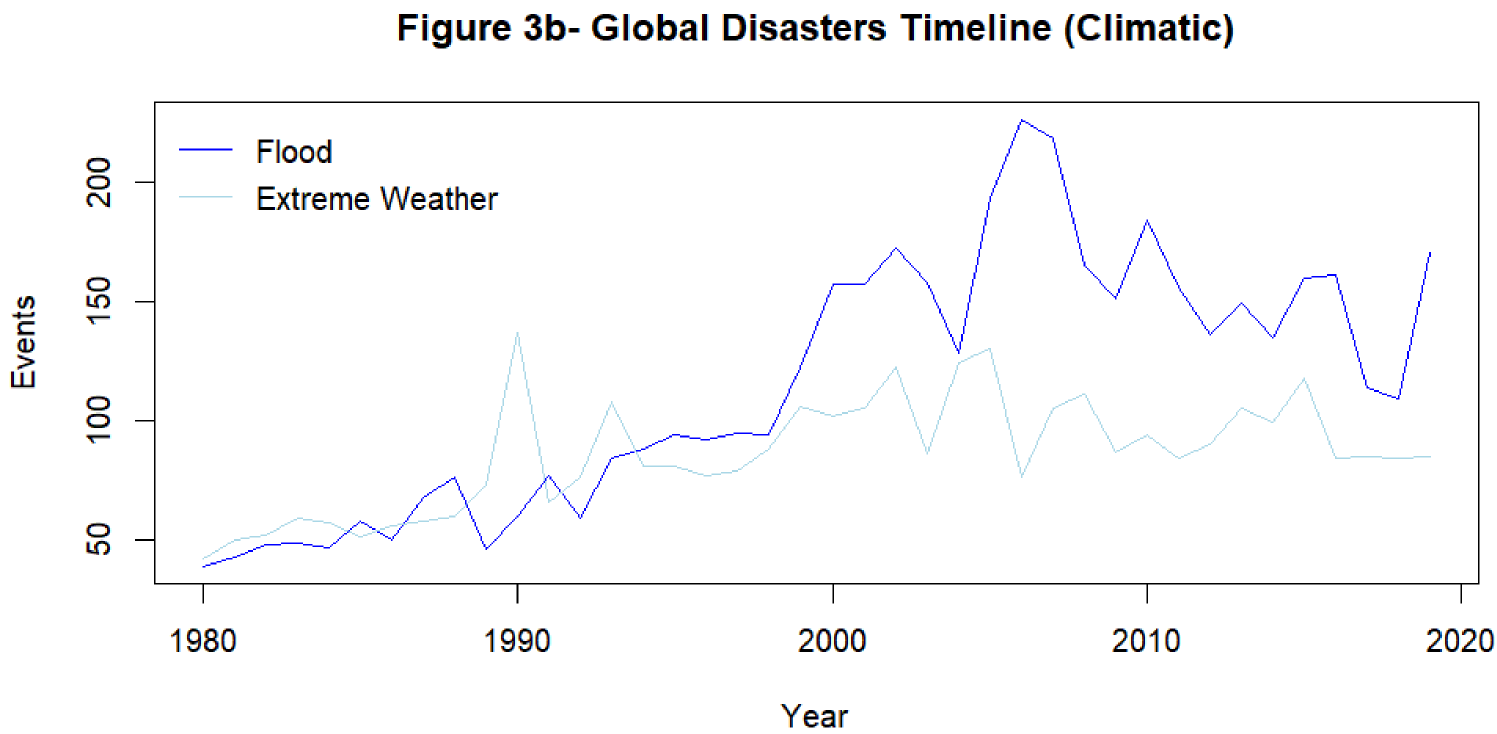


Figure 3b- Global Disasters Timeline (Climatic)
Flood
Extreme Weather
Events
50
100
150
200
1980
1990
2000
2010
2020
Year

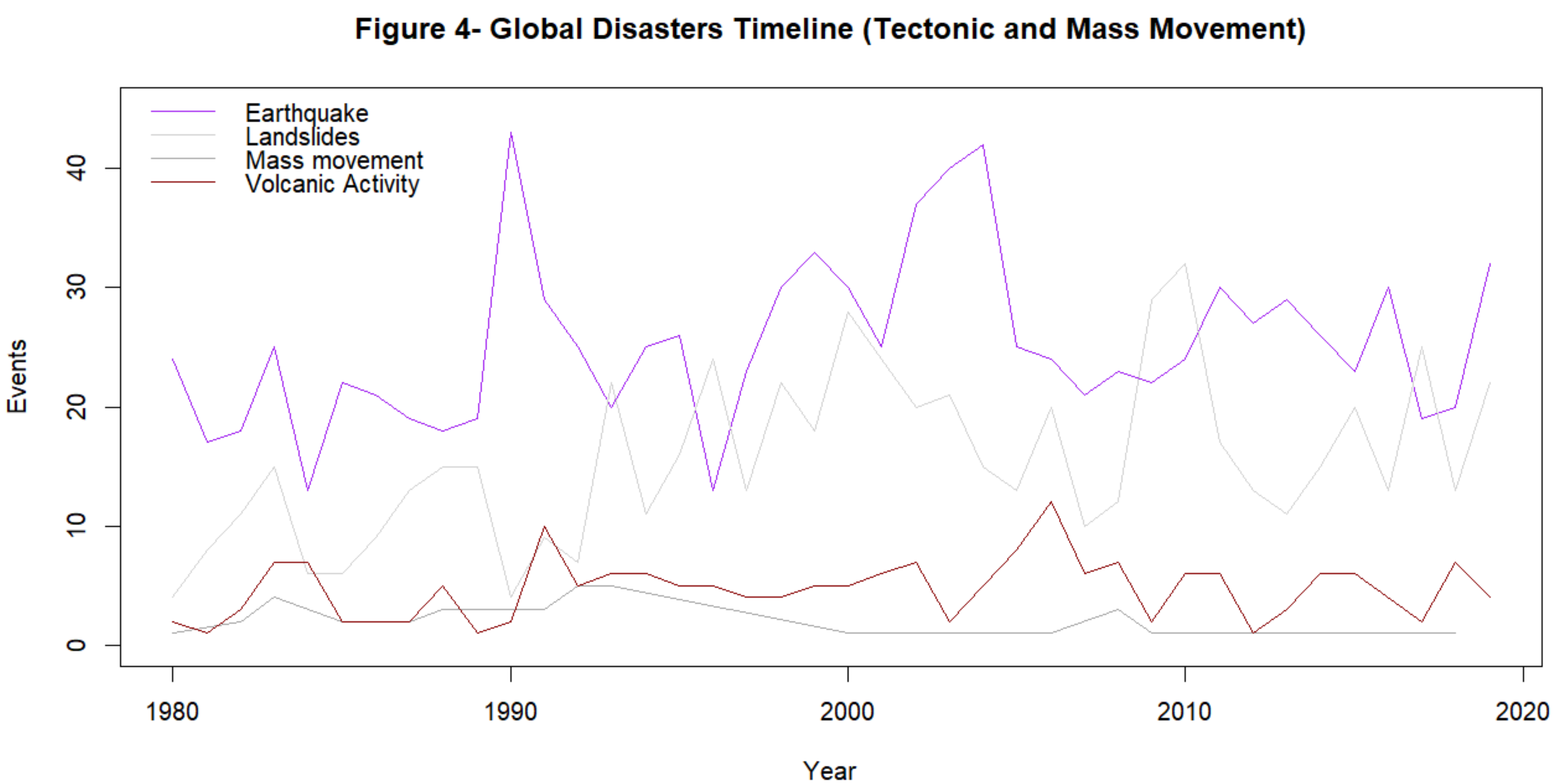
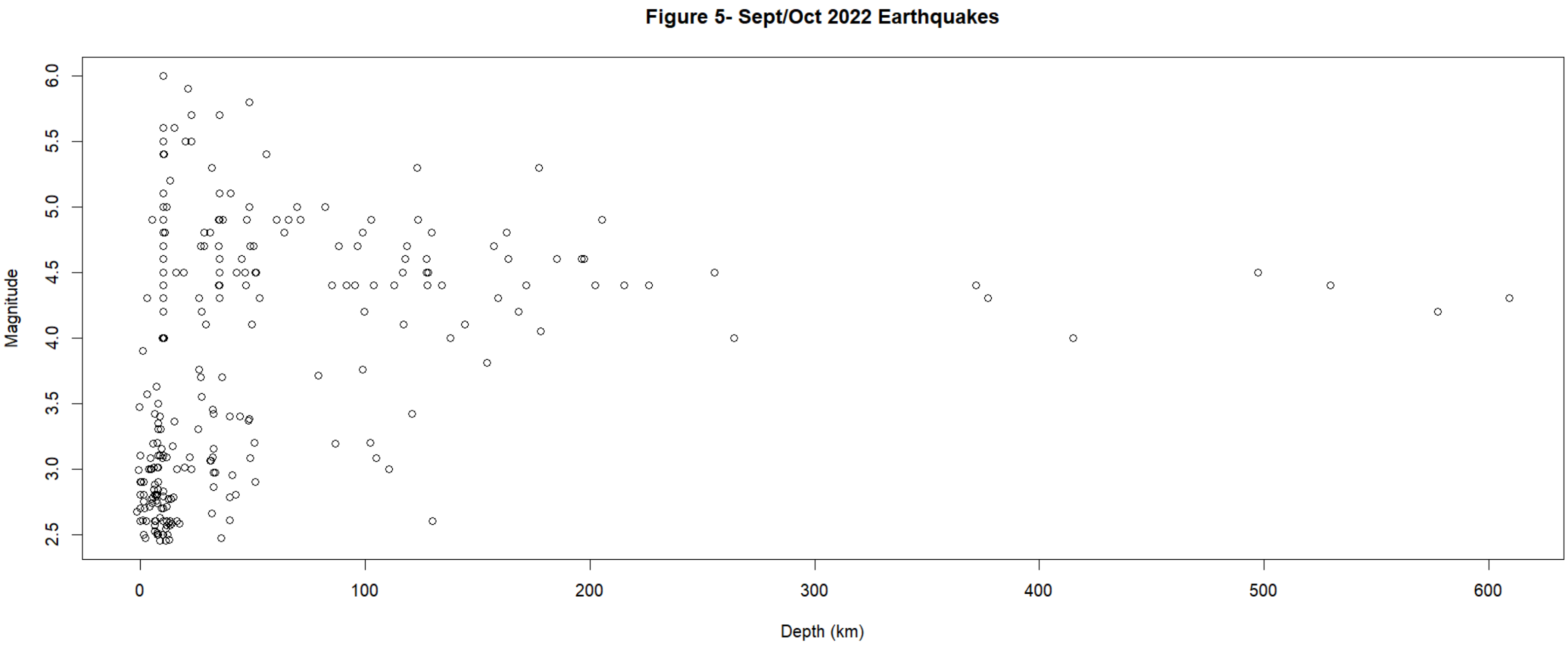
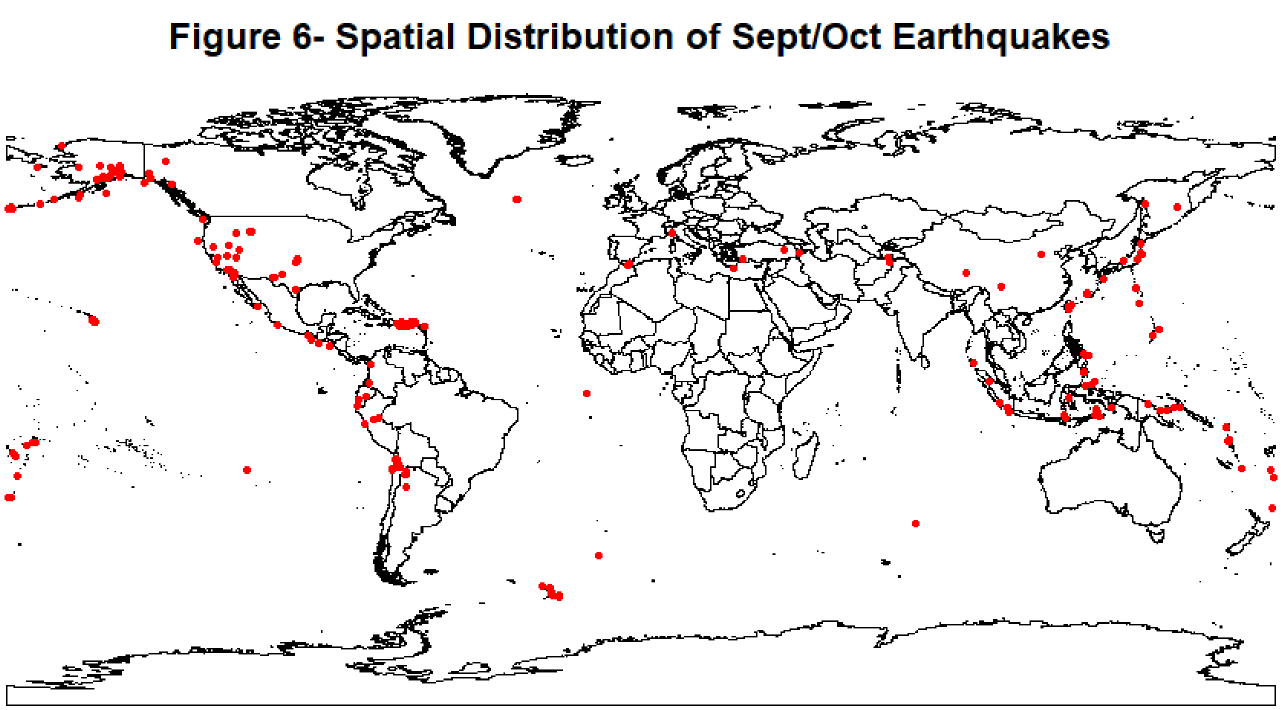


Figure 4- Global Disasters Timeline (Tectonic and Mass Movement)
Earthquake
Landslides
Mass movement
Volcanic Activity
Events
0
10
20
30
40
1980
1990
2000
2010
2020
Year

There is a high concentration of earthquakes that are both shallow and low magnitude, alongside a smaller number of shallow, higher-magnitude events. Deeper earthquakes tend to have a slightly above average magnitude (between magnitudes 3.5 and 5) however are rarely of very high magnitude (Figure 5). The spatial distribution of earthquakes is displayed in Figure 6.





The GIS analysis reveals that a number of populated areas were potentially exposed to earthquakes throughout the time period studied, of which, five case study locations were chosen (Figures 7-11). The figures show the proximity of various populated places to earthquakes events of differing magnitudes.

Figure 7- USA West Coast

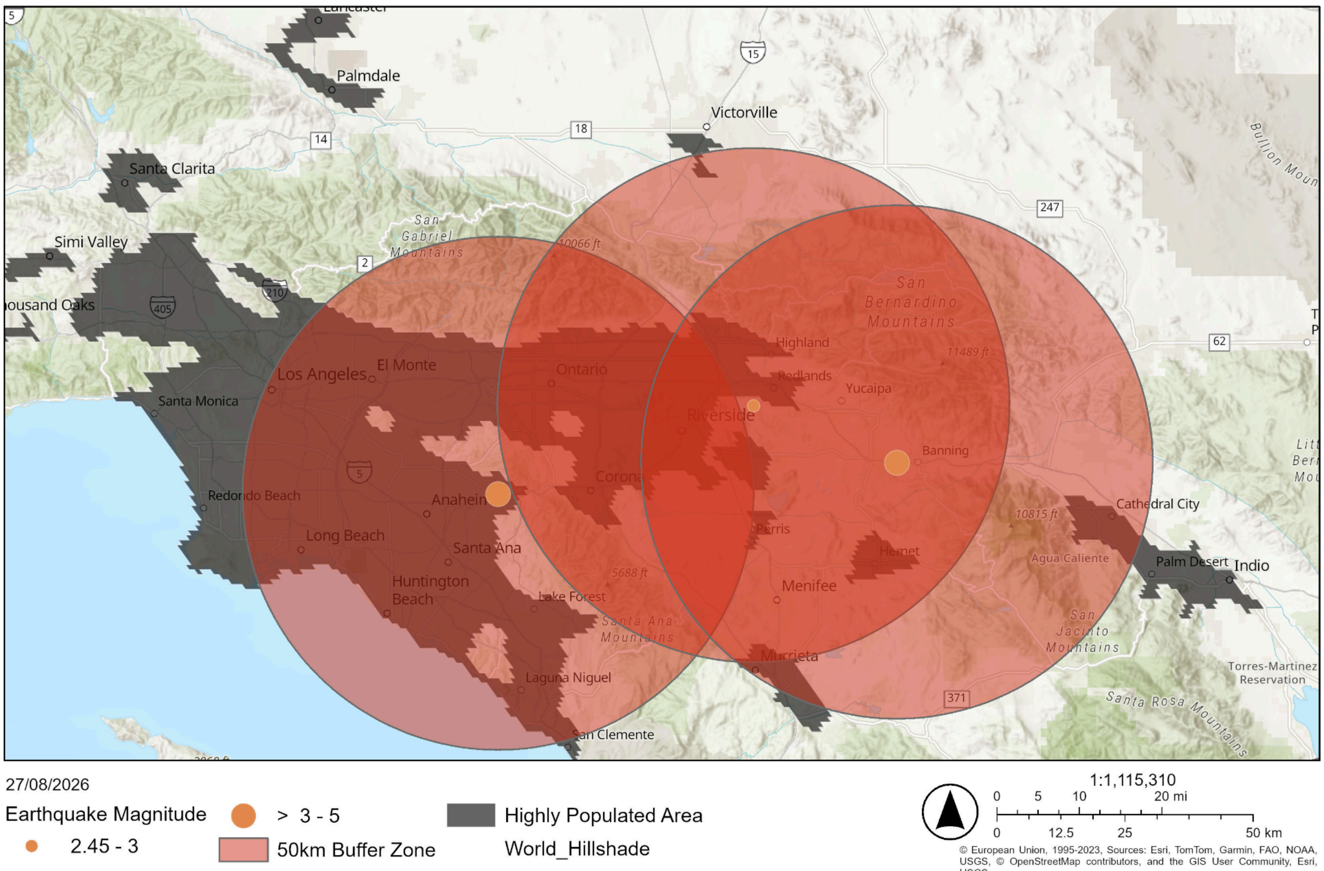
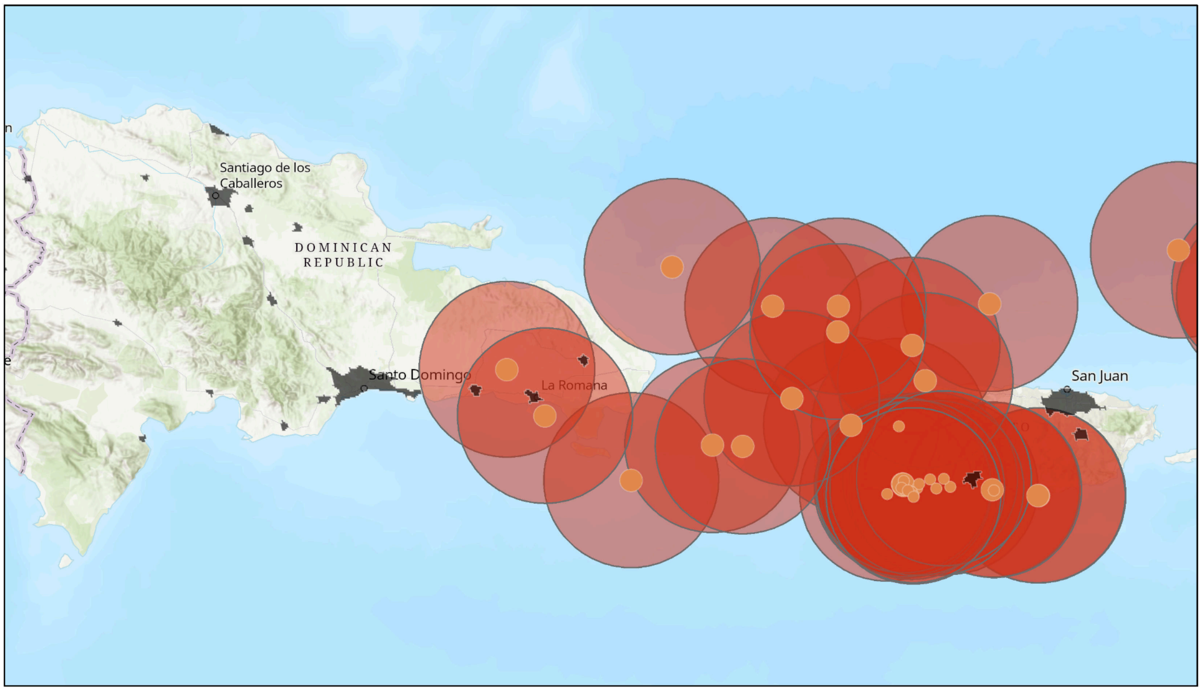


Figure 8- Dominican Republic and Puerto Rico

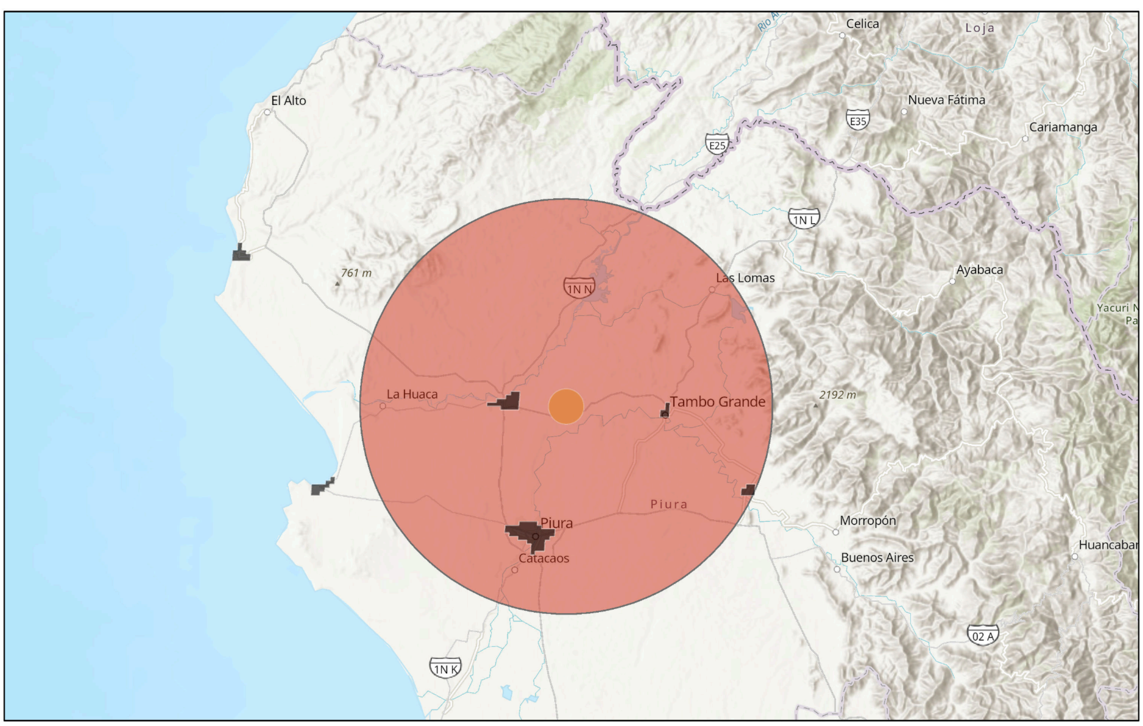


27/08/2026

Highly Populated Area
Earthquake Magnitude
2.45 - 3
> 3 - 5
50km Buffer Zone
World_Hillshade

1:2,584,556
0 20 40 80 mi
0 30 60 120 km

Figure 9- Peru, West Coast

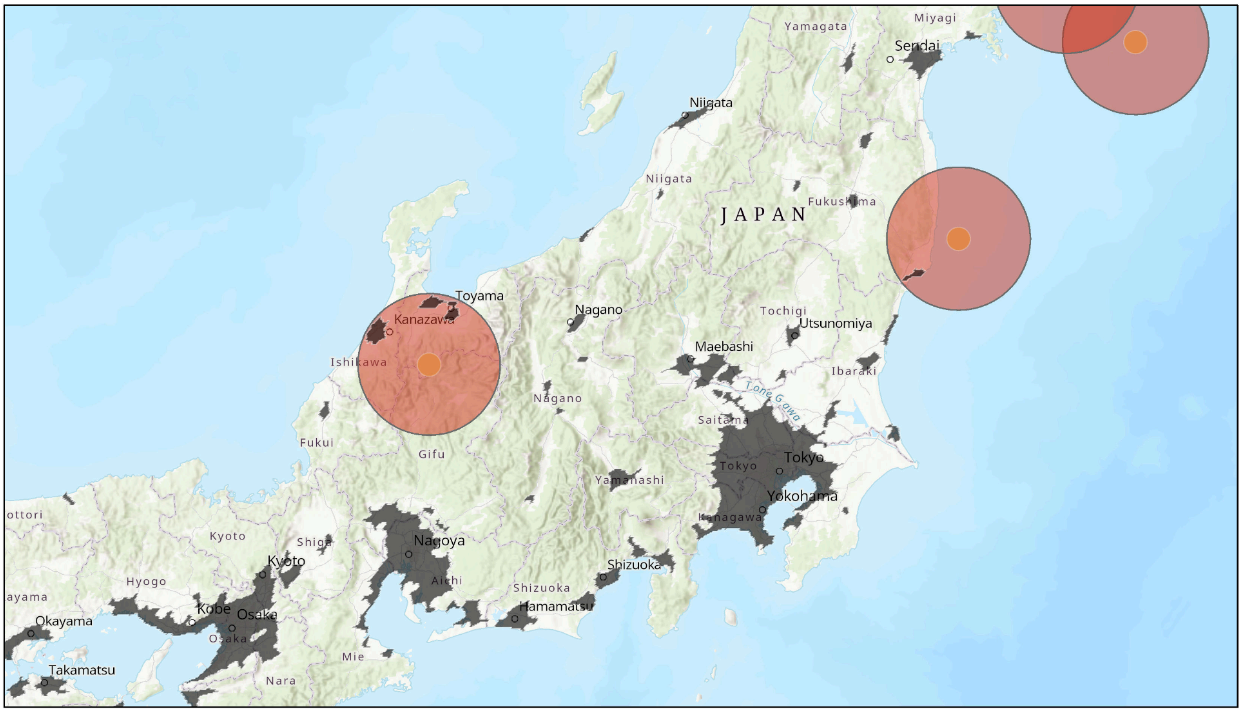


27/08/2026

Highly Populated Area
Earthquake Magnitude
> 5 - 6
50km Buffer Zone
World_Hillshade

1:1,086,863
0 5 10 20 mi
0 15 30 60 km

Figure 10- Honshu Island, Japan



27/08/2026

Highly Populated Area 50km Buffer Zone

Earthquake Magnitude World_Hillshade

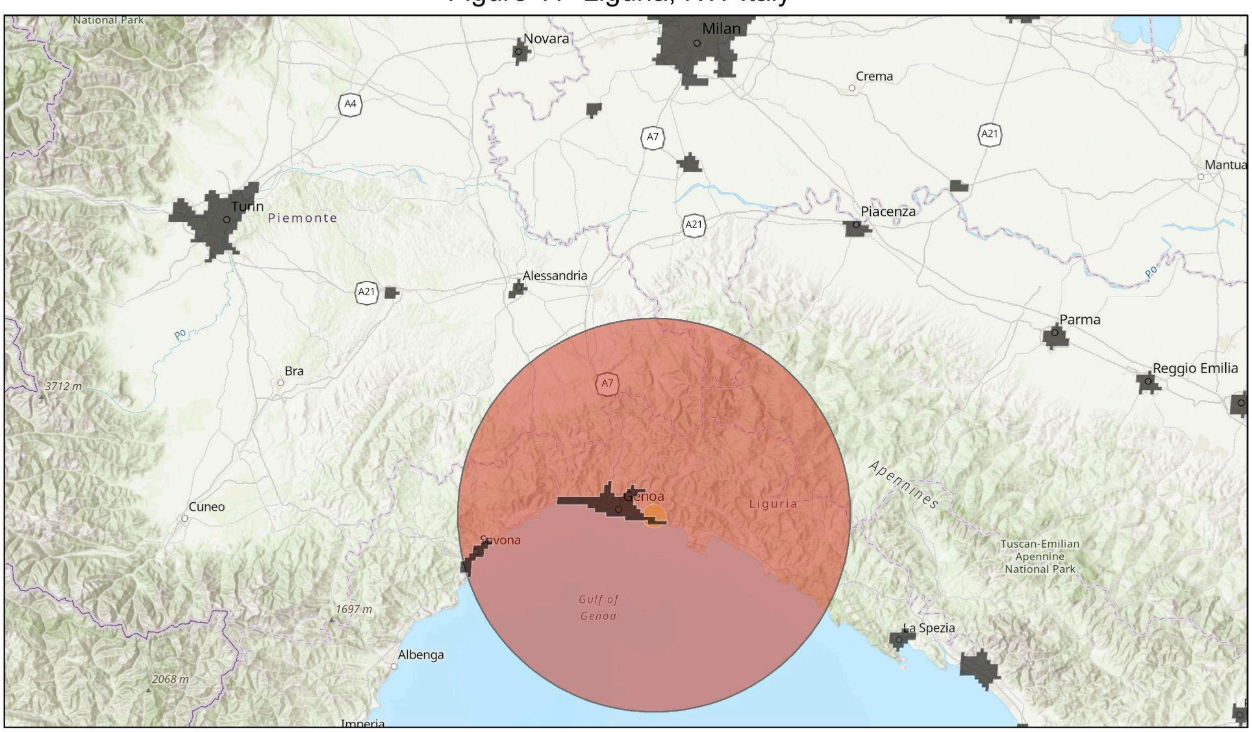
> 3 - 5

1:3,879,127

0 25 50 100 mi

0 40 80 160 km

Figure 11- Liguria, NW Italy



27/08/2026

Highly Populated Area 50km Buffer Zone

Earthquake Magnitude World_Hillshade

> 3 - 5

1:1,594,173

0 5 10 20 mi

0 15 30 60 km

Discussion

The increasing frequency of recorded climatic disasters is consistent with the broader evidence that climate change is affecting the frequency and severity of some climate-related hazards. However, attribution of the trends observed here to climate change is beyond the scope of this analysis. Figure 4 shows no trend of change over time and this is because tectonic hazards are largely unaffected by anthropogenic activities. Within this dataset, earthquake depth does not appear to have a strong relationship with magnitude. Although the maps indicate numerous populated areas within the defined exposure zones, the majority are low magnitude events. The earthquake events shown on Figures 7,8,10 and 11 are all between 2.45 and 5.0. At the lower end of this magnitude spectrum earthquakes can often not be felt, and towards the upper end damage is usually only minor. However, the earthquake event shown on Figure 9 has a magnitude of between >5.0 and 6. This means that it is likely to be strongly felt and moderate damage is probable, especially due to the close proximity of some populated areas to the epicenter.

Limitations

There are a number of simplifications used within this study. Firstly, using a 50km buffer zone to identify potentially exposed places. This is because the impact from an earthquake won't necessarily stop at, or in some cases reach, as far as this arbitrarily defined boundary. Another simplification is using populated places as the only proxy for exposure. The vulnerability, value and number of properties/infrastructure at risk aren't considered. As a result, this analysis only measures the potentially exposed locations instead of actual financial exposure. Finally, the magnitude of an earthquake isn't directly proportionate to the ground shaking experienced. An actual catastrophe model would include further measures such as Peak Ground Acceleration (PGA).

A significant limitation of the project is data availability. The earthquake dataset is very limited in terms of its duration only being a 2 and a half week period. Further data on metrics such as the monetary value of assets at risk or the vulnerability of the buildings and infrastructure would have allowed this project to go a step further and ascertain a prediction of the exposure value from some of these earthquake events. Another key weakness of this project is that unlike a commercial CAT model, the risk modelling is not probabilistic. Therefore, this project should be interpreted as an earthquake hazard and exposure assessment instead of a full catastrophe risk model.

Conclusions

This project demonstrates a simplified hazard-to-exposure workflow using R and GIS. The analysis identified patterns in recorded disaster and earthquake activity before using spatial analysis to identify populated areas potentially exposed to recorded earthquakes within a 50 km threshold. While the analysis does not incorporate vulnerability, financial exposure or probabilistic event modelling, it demonstrates several fundamental components of catastrophe risk analysis, including data preparation, hazard analysis, spatial processing and hazard-exposure matching. Understanding hazard frequency, spatial distribution and potential exposure forms an important component of catastrophe risk assessment and can contribute to insurance pricing and risk management decisions.

Appendix

```
install.packages("dplyr")
install.packages("ggplot2")
install.packages("tidyverse")
install.packages("sf")
install.packages("spdep")
library(dplyr)
library(ggplot2)
library(tidyverse)
library(sf)
library(spdep)

countries <- st_read("C:/Users/tobyj/OneDrive/Documents/UoN Datasets/Hazards repository/ne_10m_admin-0_countries")
populated_places <- st_read("C:/Users/tobyj/OneDrive/Documents/UoN Datasets/Hazards repository/ne_110m_populated_places")
earthquakes <- X2_5_week_csv

colSums(is.na(earthquakes))
colSums(is.na(disasters))
colSums(is.na(countries))
sum(duplicated(earthquakes))
sum(duplicated(disasters))
sum(duplicated(countries))
str(earthquakes)
str(disasters)
str(countries)
summary(earthquakes)
summary(disasters)
```

```
summary(countries)

all_disasters <- subset(disasters, type == "All disasters")
disasters_drought <- subset(disasters, type == "Drought")
disasters_earthquake <- subset(disasters, type == "Earthquake")
disasters_extreme_temperature <- subset(disasters, type == "Extreme temperature")
disasters_extreme_weather <- subset(disasters, type == "Extreme weather")
disasters_flood<- subset(disasters, type == "Flood")
disasters_landslide <- subset(disasters, type == "Landslide")
disasters_mass_movement_dry <- subset(disasters, type == "Mass movement (dry)")
disasters_volcanic_activity <- subset(disasters, type == "Volcanic activity")
disasters_wildfire <- subset(disasters, type == "Wildfire")

plot(all_disasters$year, all_disasters$events, type="l", xlab="Year", ylab="Events", main="Figure
2- Global Disasters Timeline", col="black")

plot(disasters_drought$year, disasters_drought$events, type="l", xlab="Year", ylab="Events",
main="Figure 3- Global Disasters Timeline (Climatic)",ylim=c(0,230), col="brown")
lines(disasters_extreme_temperature$year, disasters_extreme_temperature$events, type="l",
col="orange")
lines(disasters_extreme_weather$year, disasters_extreme_weather$events, type="l",
col="lightblue")
lines(disasters_flood$year,disasters_flood$events, type="l", col="blue")
lines(disasters_wildfire$year,disasters_wildfire$events, type="l", col="red")
legend("topleft",legend=c("Drought","Extreme Temperature","Extreme
Weather","Flood","Wildfire"),lty = 1, bty="n", col=c("brown","orange","lightblue","blue","red"))

plot(disasters_earthquake$year, disasters_earthquake$events, type="l", xlab="Year",
ylab="Events", main="Figure 4- Global Disasters Timeline (Tectonic and Mass Movement)",
ylim=c(0,45), col="purple")
lines(disasters_landslide$year,disasters_landslide$events, type="l", col="lightgrey")
lines(disasters_mass_movement_dry$year,disasters_mass_movement_dry$events, type="l",
col="darkgrey")
lines(disasters_volcanic_activity$year,disasters_volcanic_activity$events, type="l",
col="darkred")
legend("topleft",legend=c("Earthquake","Landslides","Mass movement","Volcanic Activity"), lty =
1, bty="n", col=c("purple","lightgrey","darkgrey","darkred"))

plot(disasters_drought$year, disasters_drought$events, type="l", xlab="Year", ylab="Events",
main="Figure 3a- Global Disasters Timeline (Climatic)", ylim=c(0,52), col="brown")
lines(disasters_extreme_temperature$year, disasters_extreme_temperature$events, type="l",
col="orange")
lines(disasters_wildfire$year,disasters_wildfire$events, type="l", col="red")
```

```r
legend("topleft",legend=c("Drought","Extreme Temperature","Wildfire"),lty = 1, bty="n",
col=c("brown","orange","red"))

plot(disasters_flood$year, disasters_flood$events, type="l", xlab="Year", ylab="Events",
main="Figure 3b- Global Disasters Timeline (Climatic)", col="blue")
lines(disasters_extreme_weather$year, disasters_extreme_weather$events, type="l",
col="lightblue")
legend("topleft",legend=c("Flood","Extreme Weather"),lty = 1, bty="n", col=c("blue","lightblue"))

disaster_totals <- aggregate(events ~ type, data = disasters, sum)
disaster_totals <- disaster_totals[-c(1, 7), ]

barplot(disaster_totals$events,names.arg = c("Drought","Earthquake","Ext Temp","Ext
Weather","Flood","Landslide","Mass Movement","Volcanic","Wildfire"),xlab = "Type of
Disaster",col=c("brown","purple","orange","lightblue","blue","lightgrey","darkgrey","darkred","red"
),ylab = "Events",main = "Figure 1- Frequency of Disasters")

plot(earthquakes$depth,earthquakes$mag, xlab="Depth (km)",ylab="Magnitude", main="Figure
5- Sept/Oct 2022 Earthquakes")

par(mar = c(2, 2, 2, 2))

earthquakes_sf <- st_as_sf(earthquakes,coords = c("longitude", "latitude"),crs = 4326)
plot(st_geometry(earthquakes_sf))
plot(st_geometry(countries))
plot(st_geometry(countries),col = "white",border = "black", main="Figure 6- Spatial Distribution
of Sept/Oct Earthquakes")
plot(st_geometry(earthquakes_sf),add = TRUE,pch = 16,cex = 0.5,col="red")
```